

COMICCRAFT – AI COMIC STORY CREATOR

Project Testing Phase

Project Type: Web-based AI Comic Story Generator

Testing Basis: Functional workflow, FastAPI routes, AI generation, frontend templates, comic preview and PDF export described in the project document.

1. Testing Overview

The testing phase verifies that the ComicCraft application performs its intended end-to-end workflow: accepting comic preferences, generating a structured story outline, producing narration and dialogue, creating panel illustrations, assembling the comic, displaying a preview, and exporting the completed comic as a PDF.

The project document identifies the major components and routes that form the basis of the testing activities. Testing is therefore organized around the individual AI functions, backend routes, frontend pages, comic layout and export process.

2. Testing Objectives

- Verify that the ComicCraft homepage accepts the required user inputs.
- Verify that Gemini Flash generates a five-panel outline.
- Verify that Gemini Pro generates narration and dialogue.
- Verify that Stable Diffusion produces comic-style panel illustrations.
- Verify that generated images are saved in the expected panel location.
- Verify that story content and images are correctly combined into comic panels.
- Verify that Jinja2 templates display generated comic information correctly.
- Verify that FastAPI routes execute the intended workflow.
- Verify that the final comic can be exported as a multi-page PDF.
- Verify that the export-success page is displayed after successful PDF creation.

3. Testing Scope

Area	Testing Focus
User Input	Story prompt, character name, setting, tone and art style fields
Gemini Flash	Five-panel outline structure and panel information
Gemini Pro	Narration and dialogue generation
Stable Diffusion	Comic illustration generation and image saving
FastAPI	Routes, request handling and workflow integration
Jinja2 Frontend	Input, preview and export-success pages
Comic Layout	Matching generated text with corresponding images
PDF Export	Multi-page comic creation and stored export

4. Test Environment

- Python development environment with the project's required packages.
- FastAPI application served using Uvicorn.
- Configured Google Gemini API access.
- Configured Hugging Face access for the image-generation workflow.
- Stable Diffusion and Diffusers environment for illustration generation.
- Jinja2 templates for browser-based testing.
- FPDF for PDF export testing.
- Local application access through the FastAPI server and /docs endpoint.

5. Functional Test Cases

Test ID	Test Scenario	Expected Result
TC-01	Open ComicCraft homepage	Homepage and comic input form are displayed.
TC-02	Enter story prompt and preferences	Entered character, setting, tone and art style values are accepted.
TC-03	Generate comic outline	Gemini Flash produces a five-panel structured outline.
TC-04	Generate story text	Gemini Pro produces narration and dialogue.
TC-05	Generate panel image	Stable Diffusion produces a comic-style illustration.
TC-06	Save panel image	Generated image is stored under the panel output location.
TC-07	Build comic layout	Each generated image is matched with its story information.
TC-08	Display comic preview	Preview page shows titles, images, descriptions and text.
TC-09	Export comic	FPDF creates the multi-page comic PDF.
TC-10	Show export success	Export-success page confirms completion.

6. Backend Route Testing

Route	Testing Purpose
/	Verify homepage rendering and input form.
/generate	Verify form submission and complete AI comic-generation workflow.
/generate-comic/json	Verify JSON request handling and generated layout/PDF information.
/export-success	Verify successful export confirmation page.
/test-image	Verify direct image-generation functionality.

7. AI Component Testing

7.1 Gemini Flash Testing

Check that the outline-generation function returns the planned five-panel structure and that each panel provides the information required for subsequent story and image generation.

7.2 Gemini Pro Testing

Check that the story-generation function produces narration and dialogue that can be passed into the comic layout.

7.3 Stable Diffusion Testing

Check that the image-generation function accepts the generated image prompts, creates comic-style illustrations, and saves the results in the expected static/panels location.

8. Frontend Testing

- Check that index.html displays all planned input fields.
- Check that user-entered values are passed to the backend.
- Check that comic_preview.html displays panel title, image, scene description and captions/narration.
- Check that generated panel information is rendered through Jinja2.

- Check that `export_success.html` is displayed after a successful export.

9. Comic Layout and PDF Testing

The layout-building stage is tested by checking that each generated image corresponds to the appropriate story panel and that the resulting comic contains the intended title, description, caption/narration and illustration.

The PDF export stage is tested by checking that FPDF produces a multi-page document, that the generated comic information is included, and that the resulting file is stored under the static/exports location.

10. Integration Testing

Start from the homepage and submit a complete comic request.

Confirm that FastAPI receives the request.

Confirm that the outline-generation stage completes.

Confirm that narration/dialogue generation completes.

Confirm that image generation completes for the panels.

Confirm that images and text are combined into the comic layout.

Confirm that the preview page receives the generated data.

Confirm that PDF export completes successfully.

Confirm that the export-success page is shown.

11. Error and Exception Testing

The project document specifies error handling within the FastAPI routing workflow. Testing should therefore verify the application's behavior when an AI-generation step or request processing operation does not complete successfully, while ensuring that the normal successful workflow remains intact.

- Check behavior for incomplete or invalid form input.
- Check handling when an AI generation request fails.
- Check that a failed image-generation step does not silently produce an invalid panel.
- Check that PDF export is attempted only after the required comic content is assembled.
- Check that the application reports errors through its existing route-handling mechanism.

12. Testing Checklist

- Homepage loads correctly.
- Comic input fields work correctly.
- Five-panel outline is generated.
- Narration and dialogue are generated.
- Panel illustrations are generated.
- Images are saved correctly.
- Comic layout combines images and story content.
- Comic preview displays correctly.

- PDF export succeeds.
- Export-success page is displayed.
- FastAPI /docs endpoint is accessible during local testing.

13. Testing Outcome

The testing phase provides a structured way to verify the complete ComicCraft workflow and its major components. The test cases cover user input, AI generation, backend routing, frontend rendering, panel assembly and PDF export. Successful completion of these checks supports the intended end-to-end operation described in the project documentation.

Source basis: ComicCraft – AI Comic Story Creator using Gemini Models project document.